



Final test MAE101 Fall 2019 hieptt

Discrete Mathematics (Trường Đại học FPT)



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QN=1	Let R be the real numbers set. Find the domain, D, of the function $f(x) = \frac{710}{x^2 - 19x}$
a.	$D = \mathbb{R} \setminus \{0, 19\}$
b.	$D = \mathbb{R}$
c.	$D = \{0, 19\}$
d.	$D = \mathbb{R} \setminus \{0\}$
e.	
ANSWER:	A
MARK:	
UNIT:	1
LO:	1.1
MIX CHOICES:	Yes

QN= 2	Find the range of the function $f(x) = 2 \cos x$
a.	$[-2, 2]$
b.	$[-1, 1]$
c.	$(-1, 1)$
d.	$(1, 2]$
e.	
ANSWER:	A
MARK:	
UNIT:	1
LO:	0
MIX CHOICES:	Yes

QN=3	Determine whether f is even, odd, or neither $f(x) = \frac{7x}{10x + 2019}$
a.	Neither
b.	Even
c.	Odd
d.	
e.	
ANSWER:	A
MARK:	1
UNIT:	1
LO:	1.1
MIX CHOICES:	Yes

QN=4	Let $f(x) = \sqrt{x+1}$ and $g(x) = \sqrt{5+x}$. Find $g \circ f$
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	$(i)\sqrt{6+x}$ $(ii)\sqrt{5+\sqrt{1+x}}$ $(iii)\sqrt{\sqrt{5+x}+1}$ $(iv)6+x$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	
ANSWER:	B
MARK:	
UNIT:	1
LO:	1.1
MIX CHOICES:	Yes

QN=5	Find the vertical asymptotes of the curve $f(x) = \frac{2x^2 + x + 1}{x^2 + 2x - 3}$
a.	$x = 1$ and $x = -3$
b.	$x = 1$ and $x = 3$
c.	$x = 1$ and $y = -3$
d.	$y = 1$ and $x = 3$
e.	
ANSWER:	A
MARK:	1
UNIT:	1
LO:	1.4
MIX CHOICES:	Yes

QN=6	Let $g(x) = \begin{cases} \frac{x}{2} + 3 & \text{for } x > 2 \\ x - k & \text{for } x \leq 2 \end{cases}$ Find k such that g is continuous at $x = 2$.
a.	-2
b.	2
c.	4
d.	-4
e.	
ANSWER:	A
MARK:	1
UNIT:	1
LO:	1.5
MIX CHOICES:	Yes

QN=7	Find $\lim_{x \rightarrow 0} \frac{\tan 7x}{\tan 10x}$
a.	0.7
b.	10/7
c.	0
d.	-0.7
e.	
ANSWER:	A
MARK:	1
UNIT:	1
LO:	1.4
MIX CHOICES:	Yes

QN=8	Find an equation of the tangent line to the curve at the given point: $y = \frac{x+1}{x-2}, \quad (3, 4)$
a.	$y = -3x + 13$
b.	$y = 13 + x$
c.	$y = 3x - 13$
d.	$y = -x + 7$
e.	
ANSWER:	A
MARK:	1
UNIT:	2
LO:	2.2
MIX CHOICES:	Yes

QN=9	Let: $y = -10x^2 + x\sqrt{x} + \frac{7}{x} + 2019$ Find y' . (i) $20x + \sqrt{x} - 7\ln x + 2019x$ (ii) $-20x + \frac{3\sqrt{x}}{2} - \frac{7}{x} + 2019$ (iii) $-20x + \frac{3\sqrt{x}}{2} - \frac{7}{x}$ (iv) $-20x + \frac{3\sqrt{x}}{2} - \frac{7}{x^2}$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	
ANSWER:	D
MARK:	1
UNIT:	2
LO:	2.3

MIX CHOICES:	Yes
QN=10	Let $(L): x^3 y + xy^3 = 7x$ Find dy/dx (i) $\frac{7 - 3x^2 y - y^3}{x^3 + 3xy^2}$ (ii) $\frac{7 - 3x^2 y - y^3}{x^3 - 2xy^2}$ (iii) $\frac{7 - 3x^2 y + y^3}{x^3 - 3xy^2}$ (iv) $\frac{7 + 3x^2 y - y^3}{2xy^2 - x^3}$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	
ANSWER:	A
MARK:	1
UNIT:	3
LO:	3.1
MIX CHOICES:	yes

QN=11	Each side of a square is increasing at a rate of 7 cm/s. At what rate is the area of the square increasing when the area of the square is 25 cm ² ?
a.	70
b.	35
c.	10
d.	14
e.	350
ANSWER:	A
MARK:	1
UNIT:	2
LO:	3.4
MIX CHOICES:	Yes

QN=12	Find all numbers that satisfy the conclusion of the Mean Value Theorem $f(x) = 7x^2 - 11x - 5$, on $[0;1]$
a.	0.5
b.	1
c.	0
d.	-3
e.	

ANSWER:	A
MARK:	1
UNIT:	3
LO:	3.3
MIX CHOICES:	yes

QN=13	Find the inflection points for the function $f(x) = 2019x^{2018}$
a.	There is no inflection point
b.	(0,0)
c.	(1,2019)
d.	(-1,2019)
e.	
ANSWER:	A
MARK:	1
UNIT:	3
LO:	3.5
MIX CHOICES:	Yes

QN=14	The radius of a circular disk is given as 4 cm with a maximum error in measurement of 0.02 cm. Use differentials to estimate the maximum error in the calculated area of the disk. Let $\pi = 3.14$
a.	0.5024
b.	0.2512
c.	1.0048
d.	2.0096
e.	
ANSWER:	A
MARK:	1
UNIT:	3
LO:	3.2
MIX CHOICES:	Yes

QN=15	Use Newton's method with the specified initial approximation x_1 to find x_2 . $x^5 - 2 = 0, \quad x_1 = 1$
a.	6/5
b.	4/5
c.	-4/5
d.	5/4
e.	
ANSWER:	A
MARK:	1
UNIT:	3
LO:	3.6
MIX CHOICES:	Yes

QN=16	Find the most general anti-derivative of the function.
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	$f(x) = \frac{3x^2 + 8x + 10}{x}; \quad x > 0.$ (i) $\frac{3x^2}{2} + 10 \ln x + 8x + c$ (ii) $\frac{3x}{2} + 10 \ln x + 8x + c$ (iii) $\frac{3x^2}{2} - \frac{10}{x^2} + 8x + c$ (iv) $\frac{3x^2}{2} - 10 \ln x + 8x$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	
ANSWER:	A
MARK:	1
UNIT:	4
LO:	4.3
MIX CHOICES:	Yes

QN=17	A particle is moving with the given data. Find the position of the particle. $v(t) = 2 \sin t + \cos t, \quad s(0) = 0$
a.	$S(t) = \sin t - 2 \cos t + 2$
b.	$S(t) = 2 \sin t - \cos t + 2$
c.	$S(t) = \sin t - 2 \cos t$
d.	$S(t) = 2 \sin t + \cos t - 1$
e.	
ANSWER:	A
MARK:	1
UNIT:	4
LO:	4.3
MIX CHOICES:	Yes

QN=18	Estimate the area under the graph of $y = f(x)$ using 4 rectangles and right endpoints. $f(x) = \frac{1}{x} + 2x, \quad x \in [1, 5]$
a.	29.28
b.	22.08
c.	32.28
d.	15.28
e.	
ANSWER:	A
MARK:	1

UNIT:	4
LO:	4.1
MIX CHOICES:	Yes

QN=19	Using Simpson's Rule to approximate the given integral with the specified value of n . $\int_0^4 x^2 dx, n = 4$
a.	64/3
b.	20
c.	28
d.	64
e.	
ANSWER:	A
MARK:	1
UNIT:	4
LO:	4.5
MIX CHOICES:	Yes

QN=20	$g(x) = \int_0^x \sqrt{7t^2 + 10} dt$ Find the derivative of the function (i) $x\sqrt{7x^2 + 10}$ (ii) $\sqrt{7x^2 + 10}$ (iii) $14x\sqrt{7x^2 + 10}$ (iv) $\frac{7x}{\sqrt{7x^2 + 10}}$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	
ANSWER:	B
MARK:	1
UNIT:	4
LO:	4.2
MIX CHOICES:	Yes

QN=21	Find the average value of the function on the given interval $f(x) = x + \sqrt{x}, \quad [2, 5]$
a.	5.36
b.	16.07
c.	6.26
d.	18.79
e.	

ANSWER:	A
MARK:	1
UNIT:	4
LO:	4.4
MIX CHOICES:	Yes

QN=22	A particle moves along a line so that its velocity at time t is: $v(t) = 5t^2 - 3t - 19$ (m/s) Find the displacement of the particle during the time period $1 \leq t \leq 5$
a.	284/3
b.	71/3
c.	241/4
d.	22/3
e.	
ANSWER:	A
MARK:	1
UNIT:	4
LO:	4.4
MIX CHOICES:	Yes

QN=23	A particle moves along a line with velocity function $v(t) = -8t^2 - 10t + 19$, where t is measured in meters per second. Find the distance traveled by the particle during the time interval $t \in [3, 7]$
a.	966.67
b.	-966.67
c.	332.67
d.	-332.67
e.	
ANSWER:	A
MARK:	1
UNIT:	4
LO:	4.4
MIX CHOICES:	Yes

QN=24	Suppose $f(x)$ is differentiable, $f(3) = 5$ and $\int_0^3 f(x) dx = 7$. Find $\int_0^3 xf'(x) dx$
a.	8
b.	7
c.	5
d.	15
e.	
ANSWER:	A
MARK:	1

UNIT:	4
LO:	4.3
MIX CHOICES:	Yes

QN=25	Determine whether the integral is convergent or divergent. Evaluate it if it is convergent. $\int_0^{\infty} \frac{dx}{(5x+1)^2}$
a.	It is convergent and equals to 1/5
b.	It is convergent and equals to 1/30
c.	It is convergent and equals to -1/5
d.	It is divergent
e.	
ANSWER:	A
MARK:	1
UNIT:	4
LO:	4.6
MIX CHOICES:	Yes

QN=26	Solve $\begin{cases} x + y + z = 0 \\ 2x - y + 2z = 0 \\ x - z = 0 \end{cases}$
a.	(x,y,z) = (t,-2t,t), t in R
b.	(x,y,z) = (1,-2,1)
c.	It has only one trivial solution (x,y,z) = (0,0,0)
d.	It is inconsistent
e.	
ANSWER:	C
MARK:	1
UNIT:	5
LO:	5.2
MIX CHOICES:	Yes

QN=27	Compute the rank of each of the following matrix: $A = \begin{pmatrix} 1 & 3 & 2 \\ 3 & -1 & 1 \\ 8 & 10 & 4 \end{pmatrix}$
a.	3
b.	2
c.	1
d.	4
e.	
ANSWER:	A
MARK:	1
UNIT:	5

LO:	5.1
MIX CHOICES:	Yes

QN=28	Find all values of k for which the system has nontrivial solutions $\begin{cases} x - y + 2z = 0 \\ -x + 2y - z = 0 \\ 3x + y + kz = 0 \end{cases}$
a.	10
b.	11
c.	7
d.	0
e.	
ANSWER:	A
MARK:	1
UNIT:	5
LO:	5.2
MIX CHOICES:	Yes

QN=29	Determine the values of m such that the system of linear equations has exactly one solution $\begin{cases} x + y - z = -1 \\ x + y + 2z = 2m \\ x + my + 2z = 2 \end{cases}$
a.	$m \neq 1$
b.	$m = 1$
c.	$m = 2$
d.	$m \neq -1$
e.	
ANSWER:	A
MARK:	1
UNIT:	5
LO:	5.2
MIX CHOICES:	Yes

QN=30	$A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}, B = \begin{pmatrix} -1 & 1 \\ 3 & 2 \end{pmatrix}$ Let find: $2A^T - 3B$ (i) $\begin{bmatrix} 5 & 1 \\ 1 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 5 & 1 \\ -1 & 0 \end{bmatrix}$ (iii) $\begin{bmatrix} 5 & 5 \\ -5 & 0 \end{bmatrix}$ (iv) $\begin{bmatrix} 5 & 1 \\ 1 & 3 \end{bmatrix}$
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a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	
ANSWER:	B
MARK:	1
UNIT:	6
LO:	6.1
MIX CHOICES:	Yes

QN=31	$A = \begin{pmatrix} 3 & 1 & 2 \\ 4 & 8 & 1 \\ 1 & 1 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 3 & 0 & -6 \\ 1 & 0 & 3 & 0 \end{pmatrix}$ Let A and B Find the (2,3) – entry of A.B
a.	11
b.	28
c.	4
d.	7
e.	
ANSWER:	A
MARK:	1
UNIT:	6
LO:	6.1
MIX CHOICES:	Yes

QN=32	Find the inverse of the following matrix. $\begin{pmatrix} 2 & 1 \\ 3 & -4 \end{pmatrix}$ (i) $\begin{bmatrix} -4 & 2 \\ 1 & 3 \end{bmatrix}$ (ii) $\frac{-1}{11} \begin{bmatrix} -4 & 2 \\ 1 & 3 \end{bmatrix}$ (iii) $\frac{1}{11} \begin{bmatrix} -4 & 2 \\ 1 & 3 \end{bmatrix}$ (iv) $\frac{1}{11} \begin{bmatrix} 4 & 1 \\ 3 & -2 \end{bmatrix}$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	
ANSWER:	D
MARK:	
UNIT:	7
LO:	7.2

MIX CHOICES:	Yes
QN=33	$A^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 3 & 0 & 2 \\ -1 & 4 & 0 \end{pmatrix}$ Given $XA = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 1 & 5 \end{pmatrix}$. Find a matrix X such that (i) $\begin{bmatrix} 8 & -5 & 5 \\ 1 & 17 & 5 \end{bmatrix}$ (ii) $\begin{bmatrix} 8 & -5 & 5 \\ 1 & 17 & 10 \end{bmatrix}$ (iii) $\begin{bmatrix} 8 & -5 & 4 \\ 1 & 17 & 5 \end{bmatrix}$ (iv) $\begin{bmatrix} 8 & -5 & 5 \\ 4 & 17 & 5 \end{bmatrix}$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	
ANSWER:	A
MARK:	1
UNIT:	6
LO:	6.2
MIX CHOICES:	Yes

QN=34	Find A if $(I + 2A^T)^{-1} = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix}$ (i) $\begin{bmatrix} 1 & -1 \\ 3 & -3 \end{bmatrix}$ (ii) $\frac{-1}{2} \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}$ (iii) $\frac{1}{2} \begin{bmatrix} 1 & 3 \\ -1 & -3 \end{bmatrix}$ (iv) $\frac{3}{2} \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	

ANSWER:	C
MARK:	1
UNIT:	6
LO:	6.2
MIX CHOICES:	Yes

QN=35	Evaluate the determinant $\begin{vmatrix} 3 & 2 & 1 \\ 4 & 5 & 6 \\ 2 & 3 & 1 \end{vmatrix}$
a.	-21
b.	-11
c.	21
d.	11
e.	
ANSWER:	A
MARK:	1
UNIT:	7
LO:	7.1
MIX CHOICES:	Yes

QN=36	$A = \begin{pmatrix} 3 & * & * & * \\ 0 & -1 & * & * \\ 0 & 0 & 2 & * \\ 0 & 0 & 0 & -1 \end{pmatrix}$ Let $\quad$ Find: $ (5A)^{-1} $
a.	1/3750
b.	3750
c.	625/6
d.	1/30
e.	
ANSWER:	A
MARK:	1
UNIT:	7
LO:	7.2
MIX CHOICES:	yes

QN=37	Let A and B be square matrices of order 4 such that $ A = 3$ and $ B = -4$. Find $ 3AB $
a.	-972
b.	-36
c.	972
d.	36
e.	
ANSWER:	A
MARK:	1

UNIT:	7
LO:	7.2
MIX CHOICES:	Yes

QN=38	Find all values of m for which the matrix is invertible $\begin{pmatrix} m & 1 & 3 \\ 1 & 3 & 2 \\ 1 & 4 & 5 \end{pmatrix}$
a.	$m \neq 0$
b.	$m = 0$
c.	$m = 2$
d.	$m \neq 2$
e.	
ANSWER:	A
MARK:	1
UNIT:	7
LO:	7.2
MIX CHOICES:	Yes

QN=39	Find the eigenvalues of the matrix $\begin{pmatrix} 3 & -2 \\ -3 & 2 \end{pmatrix}$
a.	0 and 5
b.	1 and 6
c.	3 and -2
d.	1 and -6
e.	
ANSWER:	A
MARK:	1
UNIT:	7
LO:	7.4
MIX CHOICES:	Yes

QN=40	Find the characteristic polynomial of the matrix $\begin{pmatrix} 5 & 4 \\ -2 & 1 \end{pmatrix}$ (i) $-x^2 - 6x + 13$ (ii) $x^2 + 6x + 13$ (iii) $x^2 - 6x - 8$ (iv) $x^2 - 6x + 13$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	
ANSWER:	D
MARK:	1

UNIT:	7
LO:	7.4
MIX CHOICES:	Yes

QN=41	$A = \begin{pmatrix} 5 & 0 & 2 & 4 \\ 1 & 0 & 1 & -3 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & -4 \end{pmatrix}$ Find the determinant of the matrix
a.	22
b.	-22
c.	30
d.	-30
e.	
ANSWER:	A
MARK:	1
UNIT:	7
LO:	7.1
MIX CHOICES:	Yes

QN=42	Let $x = (7, 6, 14), u = (0, 1, 4), v = (-1, 1, 2)$ and $w = (3, 1, 2)$ in R^3. Find scalars a, b and c such that $x = au + bv + cw$
a.	(a,b,c) = (1,2,3)
b.	(a,b,c) = (4,2,3)
c.	(a,b,c) = (-1,2,3)
d.	(a,b,c) = (3,1,-1)
e.	
ANSWER:	A
MARK:	1
UNIT:	7
LO:	0
MIX CHOICES:	yes

QN=43	Determine whether the set S is linearly independent or linearly dependent (i) $S = \{(1, -3), (-3, 9)\}$ (ii) $S = \{(-1, 2), (3, 1), (5, 7)\}$
a.	(i) and (ii) are linearly dependent
b.	(i) and (ii) are linearly independent
c.	(i) is linearly independent and (ii) is linearly dependent
d.	(i) is linearly dependent and (ii) is linearly independent
e.	
ANSWER:	A
MARK:	1
UNIT:	9
LO:	9.2
MIX CHOICES:	Yes

QN=44	For which values of k is set linearly dependent? $S = \{(-1, 2, 1), (k, 4, 0), (3, 1, 1)\}$
a.	-16
b.	16
c.	$k \neq 16$
d.	$k \neq -16$
e.	
ANSWER:	A
MARK:	1
UNIT:	9
LO:	9.2
MIX CHOICES:	Yes

QN=45	Find all values of m such that the set S is a basis of R^3 $S = \{(1, -1, 1), (3, 1, 0), (-2, 1, m)\}$
a.	$m \neq -5/4$
b.	$m = -5/4$
c.	$m \neq 5/2$
d.	$m = 5/4$
e.	
ANSWER:	A
MARK:	1
UNIT:	9
LO:	9.2
MIX CHOICES:	Yes

QN=46	Find a basis and the dimension of the subspace U $U = \{(s - 5t, s, t, s + 3t) \mid s, t \in R\}$
a.	A basis for U is $\{(1, 1, 0, 1), (-5, 0, 1, 3)\}$ and $\dim U = 2$
b.	A basis for U is $\{(1, 1, 0, 1), (-5, 0, 1, 3)\}$ and $\dim U = 3$
c.	A basis for U is $\{(1, 1, 0, 1), (5, 0, 1, 3)\}$ and $\dim U = 2$
d.	A basis for U is $\{(1, 1, 0, 1), (-5, 0, 1, 3)\}$ and $\dim U = 1$
e.	
ANSWER:	A
MARK:	1
UNIT:	9
LO:	9.3
MIX CHOICES:	Yes

QN=47	Find a basis and the dimension of the solution space, S, of the homogeneous system: $\begin{cases} x + 3y - 4z = 0 \\ -4x - y + 12z = 0 \end{cases}$
a.	A basis for S is $\{(32, 4, 11)\}$ and $\dim S = 1$
b.	A basis for S is $\{(22, 4, 11)\}$ and $\dim S = 1$
c.	A basis for S is $\{(32, 4, 11)\}$ and $\dim S = 3$
d.	A basis for S is $\{(32, 4, 1)\}$ and $\dim S = 3$

e.	
ANSWER:	A
MARK:	1
UNIT:	9
LO:	9.3
MIX CHOICES:	Yes

QN=48	Find a basis and dimension of the subspace $U = \text{span}\{(2,0,-3), (1,-2,1), (-2,4,5), (1,-2,2)\}$
a.	A basis for U is $\{(2,0,-3), (1,-2,1), (-2,4,5)\}$ and $\dim U = 3$
b.	A basis for U is $\{(2,0,-3), (1,-2,1)\}$ and $\dim U = 2$
c.	A basis for U is $\{(2,0,-3), (1,-2,1), (-2,4,5), (1,-2,2)\}$ and $\dim U = 4$
d.	A basis for U is $\{(2,0,-3), (1,-2,1), (-2,4,5), (1,-2,2)\}$ and $\dim U = 3$
e.	
ANSWER:	A
MARK:	1
UNIT:	9
LO:	9.3
MIX CHOICES:	Yes

QN=49	$A = \begin{pmatrix} 1 & 2 & 2 & -1 \\ 3 & 6 & 5 & 0 \\ -1 & -2 & -1 & -2 \end{pmatrix}$ Let A . Find $\dim(\text{col } A)$ and $\dim(\text{row } A)$
a.	2 and 2
b.	3 and 3
c.	2 and 3
d.	3 and 2
e.	
ANSWER:	A
MARK:	1
UNIT:	9
LO:	9.3
MIX CHOICES:	Yes

QN=50	Let $x = (13,-20,15)$; $v = (1,-2,3)$; $w = (-1,1,1)$ $U = \text{span}\{v, w\}$ Write x as a linear combination of the orthogonal basis of the subspace U.
a.	$x = 7v - 6w$
b.	$x = 19v - 6w$
c.	$x = 7v - 5w$
d.	$x = 7v + 6w$
e.	
ANSWER:	A
MARK:	1
UNIT:	9
LO:	9.4
MIX CHOICES:	Yes